(Please write your Enrollment Number) Enrollment No. 01001042022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B. Tech (Mechanical and Automation Engineering), Semester:4
(May, 2024)

Subject Code: BMA-206	**Subject: Engineering Materials**
Time :3 Hours	**Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	
	(a) Explain the term amorphous materials with suitable example.		
	(b) What is polycrystalline material? Explain with suitable diagram.		
	(c) What are point defects. Explain anyone point defect with suitable diagram.		
	(d) What is an alloy? Explain with suitable example.		
	(e) What is annealing process of heat treatment? Explain.		
	(f) What is ductile fracture? Explain with the help of suitable diagram.		
	(g) Explain Hall-Petch equation.		
	(h) What is the cause of rusting of iron? Explain.		
	UNIT-I		
Q2	What is unit cell? Explain the various type of unit cell with suitable diagrams.	**(10)**	
Q3	Define edge dislocation and twinned boundary defect with suitable diagram.	**(10)**	
	UNIT-II		
Q4	What is the need to study Iron –Carbon phase diagram? Explain the Iron-Carbon equilibrium phase diagram with neat sketch.	**(10)**	
Q5	Explain the microstructure of various phases of steel with suitable diagram.	**(10)**	
	UNIT-III		
Q6	What is the need for heat treatment of metals? Explain carburizing and nitriding processes of heat treatment.	**(10)**	
Q7	What is corrosion? Explain the various types of corrosion and methods of protection from corrosion.	**(10)**	
	UNIT-IV		
Q8	Define the term fatigue and creep. Explain the fatigue and creep failures with suitable examples	**(10)**	
Q9	Explain the strain hardening process as strengthening mechanism in solids in detail with suitable diagram.	**(10)**	